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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2025

IS2001-2 – INTERNET & WEB PROGRAMMING

(For IS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which property is used to change the background color in CSS?
A) color
B) background-color
C) bgcolor
D) background
2. How do you define inline styles in HTML?
A) Using the <tag>
B) Using the style attribute
C) Using the class attribute
D) Using the id attribute
3. What does the "alt" attribute in the tag provide?
A) A link to another image
B) The source of the image
C) The size of the image
D) A description of the image
4. Which CSS property is used to create space around elements, outside of any borders?
A) padding
B) spacing
C) margin
D) border-spacing
5. How do you specify that a table cell should span two columns in HTML?
A) rowspan="2"
B) colspan="2"
C) span="2"
D) colrow="2"
6. What does the <caption> tag in HTML do?
A) Adds a heading above a table
B) Describes an image
C) Creates a caption below an image
D) Creates a title for a video
7. Which attribute in the <form> tag specifies where to send form data?
A) target
B) method
C) action
D) enctype
8. What is the difference between the GET and POST methods in HTML forms?
A) GET is for retrieving data, POST is for sending data securely
B) GET can send large data, POST cannot
C) GET is secure, POST is not
D) GET sends data in the URL, POST sends data in the request body
9. Which of the following CSS property is used to specify whether the table cells share the common or separate border?
A) border-collapse
B) border-radius
C) border-spacing
D) border-merge

10. Which of the following property sets the background image to scroll or not to scroll with its associated element's content?
 A) background-scroll B) background-attach
 C) background-attachment D) background-allowance
11. In what way does DOM represent a webpage?
 A) Dynamic structure B) Recursive structure
 C) Hash table structure D) Tree structure
12. Why event handlers are needed in JS?
 A) Adds innerHTML page to the code B) Change the server location
 C) Performs handling of exceptions and occurrences D) Allows JavaScript code to alter the behaviour of windows
13. The function definitions in JavaScript begins with _____
 A) Identifier and Parentheses B) Return type and Identifier
 C) Return type, Function keyword, Identifier and Parentheses D) Identifier and Return type
14. The statement `a===b` in JS refers to
 A) Both a and b are equal in value, type and reference address B) Both a and b are equal in value
 C) Both a and b are equal in value and type D) There is no such statement
15. How can you access an HTML element with the id "myElement" using JavaScript?
 A) `document.getElementById("myElement")` B) `getElementById("myElement")`
 C) `element.getElementById("myElement")` D) `HTML.getElementById("myElement")`
16. Which of the below can be used to change the content of an HTML element using JavaScript?
 A) `setContent()` B) `innerHTML()`
 C) `setInnerHTML()` D) `innerHTML`
17. Why is the Virtual DOM faster than the real DOM?
 A) It uses hardware acceleration B) It minimizes direct interaction with the real DOM by batching updates
 C) It uses more efficient programming languages D) It does not require rendering components
18. What is the correct way to define a functional component in React?
 A) `function MyComponent() { return, <div>Hello!</div>; }` B) `const MyComponent = React.createComponent({ return <div>Hello!</div> });`
 C) `class MyComponent extends React.Component { render() { return <div>Hello!</div>; } }` D) `function MyComponent = () => <div>Hello!</div>;`
19. What is the primary purpose of props in React?
 A) To store component data B) To track state changes in a component
 C) To pass data from a parent component to a child component D) To trigger a re-render of the component
20. Which of the following lifecycle methods is part of the mounting phase in React?
 A) `componentWillReceiveProps()` B) `render()`
 C) `shouldComponentUpdate()` D) `componentDidUpdate()`

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Illustrate any four types of CSS selectors with appropriate examples	8	L*2	1	1
	b) Design an HTML table to display a list of five popular books in a library with columns for Sl. No., Title, Author, Genre, and Cover Image. Provide a suitable caption for the table. Span appropriate cells for books with the same genre. The table must have a border and contents must be center-aligned	8	L3	2	1

2. a) Explain the purpose of <a> tag and the target attribute. Demonstrate with examples how to create link to a file:

- i) in the same directory
- ii) one level above the current directory
- iii) inside a subdirectory (one level below)
- iv) in the root directory of the website
- v) using an absolute URL

8 L2 1 1

- b) Develop HTML form titled "University Admission Application" page that includes the following:

- i) Group the below form inputs under categories like "Applicant Details"
- ii) Include a textbox for "Full Name."
- iii) Add a radio button group for selecting "Mode of Study" (options: "Regular," "Distance Learning," "Online").
- iv) Include a dropdown menu to choose the "Preferred Course" (any 3 values).
- v) Provide a numeric input for "12th Grade Percentage" (with a minimum of 0 and a maximum of 100).
- vi) Add buttons for form submission and clearing

8 L3 2 1

3. a) Explain the different types of lists in HTML. Also develop a webpage as shown in Fig. Q3(a):

- Morning Routine
 1. Wake up at 6:00 AM
 2. Brush teeth
 3. Exercise for 30 minutes
- Work Routine
 1. Start work at 9:00 AM
 2. Attend meetings
 3. Complete tasks
- Evening Routine
 1. Eat dinner
 2. Relax and watch TV
 3. Go to bed by 10:00 PM

Fig. Q3(a)

8 L3 1 1

- b) Compare the GET and POST methods of HTML forms, highlighting their differences with suitable examples

8 L2 2 1

Unit – II

4. a) Explain with examples the different ways of declaring variables in JavaScript

8 L2 3 1

- b) Write a JavaScript program that takes a user's birth year as input, validates if it's a valid number, calculates the age, and classifies them as:

- i) "Child" if age is less than 13
- ii) "Teenager" if age is between 13 and 19
- iii) "Adult" if age is 20 or above

8 L3 4 1

5. a) Create an HTML table to display employee details. Style the table to:

- i) Alternate the background color of rows.
- ii) Highlight rows when hovered.
- iii) Apply rounded corners and a dotted border to the table and cells.

8 L3 3 1

b)	Write a JavaScript code to design a simple calculator to perform the basic arithmetic operations	8	L3	4	1
6. a)	Explain any four ways to display output in JavaScript with examples.	8	L2	3	1
b)	Write JavaScript to validate the following fields of the Login form:				
	i) Username (field cannot be empty)				
	ii) Email (valid email format validation)				
	iii) Password (minimum length of 8 characters)				
	iv) Confirm Password (must match the password entered in the Password field)	8	L3	4	1

Unit – III

7. a)	Explain the types of react components with suitable examples.	8	L2	5	1
b)	Write a React JS code to demonstrate how to locally manage a component's state and explain how it works.	8	L3	5	1
8. a)	Explain State and Props in ReactJS with suitable examples	8	L2	5	1
b)	Design a React.js application to display a User Profile Card that includes User's Name, Age, and Location. Use inline styling to style the profile, where:				
	i) The User's Name should be displayed in bold with a larger font size.				
	ii) The Age should be displayed in italic and with a green color.				
	iii) The Location should have a background color of light blue and white text.				
	iv) Apply styling to the profile card with padding and border radius	8	L3	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
